CHAPTER 17

Master Formula Sheet & Mixed Practice

Every formula of Chapters 2–15 on one place · exam traps · 14 mixed questions with answers

QUICK TRICK

Read this sheet the night before. Every row points back to its chapter. If a row feels unfamiliar, reopen that chapter.

17.1 Formula sheet

● **LEVEL 2 · EXAM**

Topic (Ch)	Formula / fact
Delays (3)	$T_t = L/R$ · $T_p = d/v$ · $RTT = 2T_p$ · BDP = R × RTT · N links, P packets: $(N + P - 1) \times L/R$
Optimal packet (3)	$p = \sqrt{(M \cdot h / (k - 1))}$
Error detection (4)	detect d: $d_{min} \geq d+1$ · correct t: 2t+1 · CRC bits = degree of G · burst ≤ r always caught · checksum receiver sum = 0xFFFF
Stop-and-wait (5)	$a = T_p/T_t$ · $\eta = 1/(1 + 2a)$ · with loss p: $(1 - p)/(1 + 2a)$
Sliding window (5)	$\eta = \min(1, W/(1 + 2a))$ · $W_{opt} = 1 + 2a$ · GBN $W \leq 2^k - 1$ · SR $W \leq 2^{k-1}$ · loss: GBN $(1-p)/(1+2ap)$, SR $1-p$
ALOHA (6)	Pure $S = Ge^{-2G}$ (max 0.184) · Slotted $S = Ge^{-G}$ (max 0.368)
CSMA/CD (6)	$L_{min} = 2 \cdot T_p \cdot B$ · $d_{max} = L \cdot v/(2B)$ · backoff $K \in \{0 \ldots 2^i - 1\} \times 512$ bit-times (cap 1023)
Ethernet (7)	frame 64 – 1518 B (payload 46 – 1500) · preamble+SFD 8 B extra · switch learns / forwards / floods
DV (8)	$D_x(y) = \min_v [c(x,v) + D_v(y)]$ · RIP max 15 hops · fixes: split horizon, poison reverse
LS (8)	flood LSAs → full map → Dijkstra ($O(V^2)$, or O(E log V) with heap)
CIDR (9)	block = 2^{32-n} · hosts = block − 2 · network = IP AND mask · block aligned to its own size
Fragmentation (10)	data/frag = $\lfloor (MTU - H)/8 \rfloor \times 8$ · offset = bytes/8 · MF = 1 except last · total length ≤ 65535 · header 20 – 60 B
Forwarding (10)	longest-prefix match · default 0.0.0.0/0 · TTL − 1 at each router
NAT (11)	out: change source (IP, port) · in: change destination · ≈ 2^{16} connections per public IP
TCP (12)	ACK = next byte expected · SYN/FIN use 1 seq · MSS = MTU − 40 · wrap rate ≤ $2^{32} \cdot 8/MSL$ · RTO = Est + 4·Dev
TCP control (13)	window = min(rwnd, cwnd) · rate ≤ W/RTT · slow start ×2 per RTT, then +1 · timeout: ssthresh = cwnd/2, cwnd = 1 · 3 dup ACKs Reno: cwnd = ssthresh
Sockets (14)	connect sends SYN · listen = passive · accept returns new socket · UDP uses sendto/recvfrom
DNS / HTTP (15)	DNS UDP 53, root → TLD → authoritative · HTTP TCP 80 · non-persistent 2 RTT per object · pipelined 3 RTT total

17.2 Ten traps that cost marks

COMMON MISTAKE

- Units: bits vs bytes, ms vs s, Mbps = 10^6 (not 2^{20}).
- T_p counted once per link; RTT = $2T_p$.
- GBN window $2^k - 1$, SR window 2^{k-1}; round W up.
- Hosts = block − 2; block must be aligned.
- Fragment offset in 8-byte units; non-last data multiple of 8; every fragment has its own header.
- Longest prefix, not first match.
- NAT changes the source on the way out, the destination on the way in.
- SYN and FIN each use one sequence number; ACK = next byte wanted.
- Reno after 3 dup ACKs: cwnd = ssthresh (not 1); timeout: cwnd = 1.
- accept() returns a new socket; connect() sends the SYN.

17.3 Mixed practice

Practice questions

● **LEVEL 2 · EXAM**

Q1. A 1500 B packet crosses 3 links of 10 Mbps, each with 1 ms propagation delay (store-and-forward, no queuing). End-to-end time?

Q2. Stop-and-wait link: 4 Mbps, frame 2000 bits, one-way propagation 2 ms. Find efficiency, window for 100% and sequence bits for GBN and SR.

Q3. Router table: 192.168.0.0/16 → A, 192.168.4.0/22 → B, 192.168.5.0/24 → C. Where do packets for 192.168.5.9, 192.168.6.1 and 192.168.9.1 go?

Q4. A company needs 6 subnets of at least 25 hosts from 10.10.0.0/24. Mask?

Q5. A 2500 B datagram (20 B header) meets MTU 800. Fragment data sizes and offsets?

● **LEVEL 3 · GATE**

Q6. NAT public IP 203.0.113.1 has entry (192.168.1.10, 5000) ↔ (203.0.113.1, 7000). A packet from 8.8.8.8:80 to 203.0.113.1:7000 arrives. What leaves the NAT?

Q7. Client SYN has ISN 700, server ISN 3000. What ACK number does the third handshake segment carry, and what is the server's ACK after the client sends 200 B of data?

Q8. Initial ssthresh = 12, cwnd starts at 1. At round 7 cwnd = 14 and 3 duplicate ACKs arrive. What is cwnd at round 10 under Reno and under Tahoe?

Q9. A CRC generator is $x^7 + x^3 + 1$. How many bits are appended, and which bursts are always detected?

Q10. Minimum frame size for CSMA/CD at 1 Gbps with a maximum span of 200 m ($v = 2\times10^8$ m/s)?

Q11. P has neighbours Q (cost 1) and R (cost 4). Q reports cost 7 to T, R reports cost 2. P's cost and next hop to T?

Q12. A page has 1 base file + 20 objects, RTT = 40 ms (ignore DNS and transmission). Time for persistent with and without pipelining?

Q13. One host opens two simultaneous TCP connections to the same server port 80. What makes the two connections different?

Q14. DNS and HTTP together: DNS needs 4 query/response pairs. How many DNS messages cross the network?

Answers

Q	Solution
1	$T_t = 12000/10^7 = 1.2$ ms per link. Total = 3 × 1.2 + 3 × 1 = **6.6 ms**.
2	$T_t = 0.5$ ms, a = 4. η = 1/9 = **11.1%**. W = 1 + 8 = **9**. GBN: $2^k - 1 \geq 9 \rightarrow$ **k = 4**. SR: $2^{k-1} \geq 9 \rightarrow$ **k = 5**.
3	192.168.5.9 matches /16, /22 (4.0 – 7.255) and /24 → **C**. 192.168.6.1 matches /16 and /22 → **B**. 192.168.9.1 matches /16 only → **A**.
4	6 subnets need 3 bits (8 subnets) → /27 → 30 hosts ≥ 25. Mask **255.255.255.224**.
5	Data 2480; per fragment $\lfloor 780/8 \rfloor \times 8 = 776$. Four fragments: data **776, 776, 776, 152**; offsets **0, 97, 194, 291**; MF = 1, 1, 1, 0.
6	Destination rewritten to **192.168.1.10:5000**; source stays 8.8.8.8:80.
7	Third segment: ACK = **3001**. Data starts at seq 701, 200 B → server ACK = **901**.
8	Rounds 1–7: 1, 2, 4, 8, 12, 13, 14. New ssthresh = 7. Reno: round 8 = 7, 9 = 8, 10 = **9**. Tahoe: round 8 = 1, 9 = 2, 10 = **4**.
9	**7** bits. All bursts of length **≤ 7**.
10	$T_p = 200/2\times10^8 = 1$ µs. L = 2 × 1 µs × 10^9 = **2000 bits = 250 B**.
11	via Q: 1 + 7 = 8; via R: 4 + 2 = 6. Cost **6 via R**.
12	Without pipelining: 2 × 40 + 20 × 40 = **880 ms**. With pipelining: 3 × 40 = **120 ms**.
13	The client's **source port** (ephemeral ports differ), giving different 4-tuples.
14	4 queries + 4 responses = **8 messages**.

Done. Re-attempt the mixed practice after two days, then solve previous-year GATE questions and tag each as in-syllabus or low-priority using Chapter 1.

AMJESH singhamjesh.github.io